Experiment 7

Title: Identify Semantic Relationships Between Words Using WordNet

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Aim

To identify **semantic relationships between words** from a given text using the WordNet dictionary.

Objectives

1. To understand the concept of semantic relationships in Natural Language Processing.
2. To study the use of WordNet for analyzing word meanings and relationships.
3. To identify relationships such as synonyms and antonyms between words in text.

Theory

1. Introduction to Semantic Relationships
Semantic relationships refer to the **meaning-based connections between words** in a language. In Natural Language Processing (NLP), understanding these relationships helps computers interpret the meaning and context of text. Examples include relationships such as synonymy, antonymy, and hypernymy.

2. WordNet Dictionary
WordNet is a widely used **lexical database for the English language**. It groups words into sets of synonyms called *synsets*, where each set represents a specific concept or meaning. WordNet also provides definitions and examples to help understand how words are used in different contexts.

3. Types of Semantic Relationships in WordNet
WordNet identifies different types of relationships between words. **Synonyms** are words with similar meanings, while **antonyms** are words with opposite meanings. It also includes **hypernyms** (general terms) and **hyponyms** (more specific terms), which help show hierarchical relationships between words.

4. Applications of WordNet in NLP
WordNet is widely used in **text analysis, information retrieval, machine translation, and question answering systems**. By identifying semantic relationships, it helps improve the understanding of language and enhances the accuracy of NLP applications.

Conclusion

In this experiment, semantic relationships between words were identified using the WordNet dictionary. WordNet helps analyze the meanings of words and their relationships with other words. By studying synonyms, antonyms, and hierarchical relations, text understanding can be improved. This technique is useful in many NLP applications. Therefore, WordNet plays an important role in semantic analysis of text.